Fine-tuning, Reinforcement Learning and Parameter-Efficient Adaptation

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What we will be talking about

1. **Naively Finetuning LLMs**
   - What is finetuning?
   - Why not (fully) finetune LLMs?
2. **Making finetuning practical through LoRA**
   - Understanding why LoRA works
   - How to use LoRA
3. **Making finetuning scalable through QLoRA**
   - Quantizing LLMs: Why and how?
   - How does it differ from LoRA?

Outline

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Section 1

Naively Finetuning LLMs

Subsection 1

What is finetuning?

Why finetune LLMs?

- **Prompting:** Changes input behavior. We guide the model with instructions and examples.
- **Fine-Tuning:** Changes model parameters. We switch the knowledge of the model.

> Reminder
>
> **Fine-Tuning (FT):** The process of further training a pre-trained model on a smaller, domain-specific dataset to adapt its general knowledge to a specialized task.

Pretraining vs Finetuning

Reminder

- Pre-training provides the model with initial, general knowledge by utilizing any and all data sources.
- Finetuning shifts existing knowledge, to specific domains using domain-specific datasets.

Example: Domain-specific tasks

Part-of-Speech (POS) Tagging

The process of assigning grammatical categories to each word in a sentence.

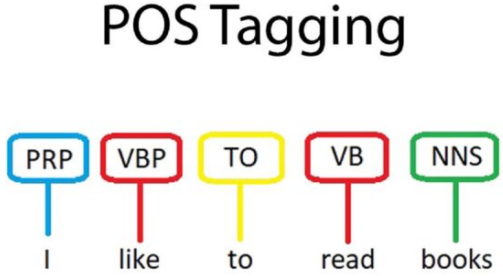

Figure: `https://byteiota.com/pos-tagging/`

Example: Domain-specific tasks

Biomedical POS Tagging

- Patterns absent or rare in general-domain text.
- Prompting probably not enough due to missing knowledge.

Word	POS Tag
Homozygous	JJ
null	JJ
mutants	NNS
die	VBP
shortly	IN
after	RB
birth	NN

Table: Example from the CRAFT dataset.

Section 1

Naively Finetuning LLMs

Subsection 2

Why not (fully) finetune LLMs?

High Resource Cost

- **Computational Power:** Full FT requires massive hardware requirements and thousands of compute hours for large models (e.g., Llama-70B).
- **Storage:** Each specialized task requires a separate, full copy of the massive foundational model.

Operational Overhead

- Training is slow.
- Deployment becomes complex, managing multiple large models for different applications.
  - A single 13B model may require 26GB of storage.
  - 10 finetuned models may thus require 260GB.
- High risk of Catastrophic Forgetting—the model forgets its general knowledge while learning the specific task.

Scalability issues

BERT

- **Model Size:** $\sim$340M Parameters
- **VRAM Requirement:** $\approx$ 8-12GB
- **Training Time:** Minutes to a few hours

Llama 70B

- **Model Size:** $\sim$70 Billion Parameters
- **VRAM Requirement:** $\approx$1000GB+
- **Training Time:** Days to Weeks (even with specialized hardware)

LLMs are too large to run full-finetuning. But do we need to?

Parameter-Efficient Fine-Tuning (PEFT)

Definition

PEFT: A set of techniques used to adapt large pre-trained LLMs to specific tasks by updating only a small subset of their parameters, rather than retraining the entire model.

- Many algorithms developed through the years (adapters, prompt/prefix tuning etc.).
- However, the LoRA family of methods has dominated the LLM space in the last few years.

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Section 2

Making finetuning practical through LoRA

Subsection 1

Understanding why LoRA works

PEFT: LoRA

- Low-Rank (LoRA) Matrix Decomposition is a specific PEFT algorithm that exploits the fact that weight updates in LLMs have a “low intrinsic rank”.
- Low rank matrices enable us to scalably represent information.

Low-rank matrices

What is a matrix?

A matrix is essentially a organized collection of data (rows and columns). For large datasets, these matrices can become incredibly complex, requiring massive storage and computational power.

Example: Data matrix

$$\begin{bmatrix} 2 & 4 & 6 & 8 & 10 \\ 1 & 2 & 3 & 4 & 5 \\ 3 & 6 & 9 & 12 & 15 \\ 4 & 8 & 12 & 16 & 20 \end{bmatrix}$$

Low-rank matrices

Matrix rank

- The 'true complexity' or the number of genuinely independent patterns within the matrix M.
- Intuitively, it tells us how many underlying 'ingredients' or fundamental directions are needed to describe the entire matrix.

Example: Cooking

Think of a restaurant with 100 dishes but only 3 main food styles: Italian, Mexican, and Vegetarian. Each dish can be described as a mix of these few styles.

Low-rank matrices

Rank Factorization

If a matrix M has a low rank k, we can approximate it (or exactly represent it) by factoring it into a product of two simpler matrices.

Example: Rank 1 matrix

$$\begin{bmatrix} 2 & 4 & 6 & 8 & 10 \\ 1 & 2 & 3 & 4 & 5 \\ 3 & 6 & 9 & 12 & 15 \\ 4 & 8 & 12 & 16 & 20 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 3 \\ 4 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \end{bmatrix}.$$

Why are LLM updates typically low-rank?

Many possible explanations

- Updates generally follow the direction of existing knowledge $\rightarrow$ the updates may often just be re-weighting of existing weights.
- Underlying ML problems are usually inherently low-rank.

- Imagine a piano with 88 keys.
- Most songs don't require arbitrary use of all 88 keys; they can be described by general patterns (e.g., melody, rhythm).
- Low-rank updates capture changes in general patterns.

Why are LLM updates typically low-rank?

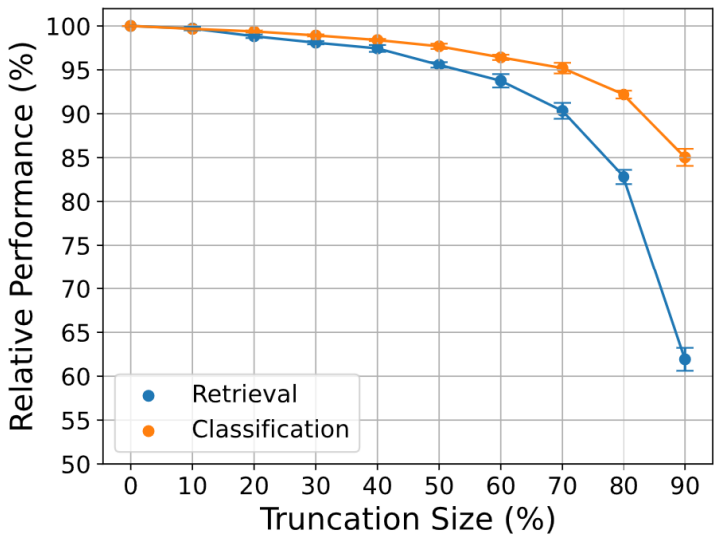

Figure: `https://aclanthology.org/2025.emnlp-main.1410.pdf`

Large embedding spaces

Modern NLP models are over-parameterized $\rightarrow$ not enough information to fill in higher ranks of weights.

LoRA: Low-rank decomposition

$$\underbrace{A_{m\times n}}_{\text{large matrix}} \approx \underbrace{U_{m\times r}}_{\text{left factors}} \underbrace{V_{r\times n}}_{\text{right factors}}, \qquad r \ll \min(m,n).$$

$$A_{10000\times 5000} \approx U_{10000\times 50} V_{50\times 5000},$$

$$\underbrace{10000\times 5000}_{50{,}000{,}000 \text{ parameters}} \longrightarrow \underbrace{10000\times 50 + 50\times 5000}_{750{,}000 \text{ parameters}}.$$

LoRA: Mathematical Foundation

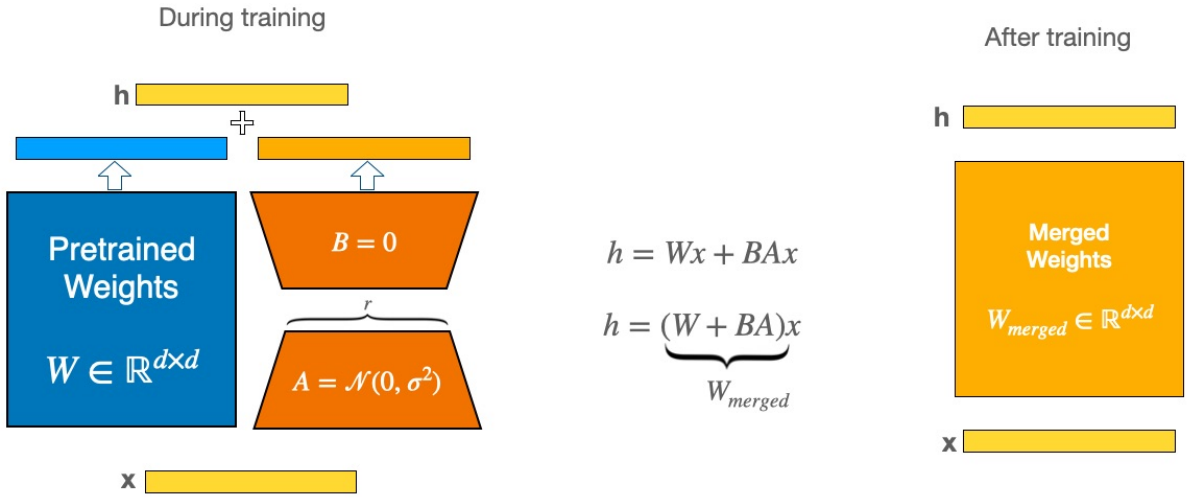

Figure: `https://huggingface.co/docs/peft/main/conceptual_guides/lora`

- $W \in \mathbb{R}^{d \times k}$ weights.
- x input, h output.
- We add BAx to the output.
- $B \in \mathbb{R}^{d \times r} \rightarrow$ zeros.
- $A \in \mathbb{R}^{r \times k} \rightarrow$ Gaussian.
- $r \ll \min(d, k)$, hence A, B are much smaller than W.

Section 2

Making finetuning practical through LoRA

Subsection 2

How to use LoRA

LoRA: Algorithm

▷ *====Initialization====* ◁

for all $W \in all\_weights$ **do**

  Set $\mathbf{W}$ to read-only ▷ *Original knowledge preserved*

for all $W_u \in target\_modules$ **do**

  Initialize $\mathbf{A}$ randomly

  Initialize $\mathbf{B} = \mathbf{0}$ ▷ *Contributes zero to the output initially*

▷ *====Backpropagation====* ◁

for all $X \in input\_embeds$ **do**

  $\mathbf{\Delta W} = \frac{\alpha}{r}\mathbf{AB}$

  $\mathbf{H} = \mathbf{W}X + \mathbf{\Delta W}X$ ▷ *Output = original + adaptation*

▷ *====End of training====* ◁

$\mathbf{W'} = \mathbf{W} + \frac{\alpha}{r}(\mathbf{AB})$ ▷ *Merge original and adaptation*

LoRA: Practical Considerations

- How many ranks do I use?
  - $r \in [1, 16]$: fast, easy, recommended for minor adjustments.
  - $r \in [16, 64]$: recommended for most domain adaptation scenarios.
  - $r \in [64, 256]$: may hit computational bottlenecks, only use for dramatic shifts in domain and knowledge.
- How do I set α?
  - Usually set to a multiple of r .
  - Very low $\alpha \rightarrow$ minimal contribution. Very high $\alpha \rightarrow$ destruction of model knowledge.
  - $\alpha = 2r$ usually used as a starting point.

LoRA: Practical Considerations

- What modules do I change?
- Start from the top of this list and go down as required:
    1. **Query (q_proj) and Value (v_proj):** Most frequently adapted; control how the model attends to input.
    2. **Key (k_proj):** Increases information retrieval capacity.
    3. **Output (o_proj):** Less common, but useful for modifying integrated attended information.
    4. **Feed-forward Networks (MLP):** Possible to adapt, but generally offer less parameter benefit compared to attention layers.

LoRA: Practical Considerations

- Should we merge the LoRA weights?

$$\mathbf{W}' = \mathbf{W} + \frac{\alpha}{r}(\mathbf{AB})$$

Yes

- Lowers GPU memory and execution time.
  - More significant if r is large.
- Deployment may be easier.

No

- You can use the same model with many finetuned adapters which are dynamically selected depending on your task.

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Section 3

Making finetuning scalable through QLoRA

Subsection 1

Quantizing LLMs: Why and how?

QLoRA: Quantization

- Quantization refers to dropping the size (not number) of the LLM parameters.
- Normally, each parameter is stored in a 16 bit floating point.
- Quantization can drop them to usually 4 bits, resulting in $\times 4$ less GPU memory and inference costs.

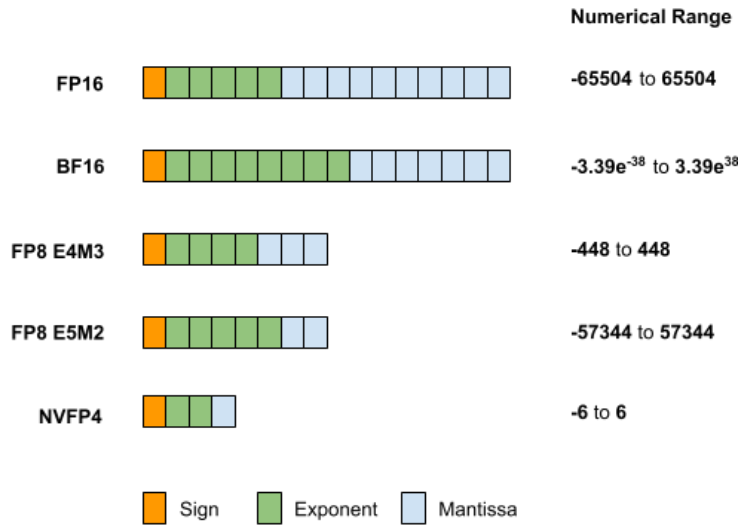
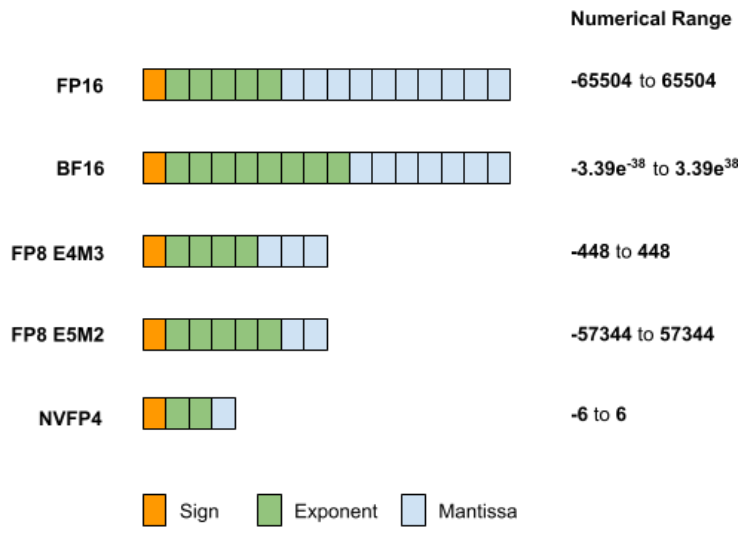


Figure: https://developer.nvidia.com/blog/model-quantization-concepts-methods-and-why-it-matters

QLoRA: Quantization

- Quantization is one of the most useful tools at your disposal.
- Allows for training and inference in consumer-grade hardware.
- We will return to this in Module 5.



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Section 3

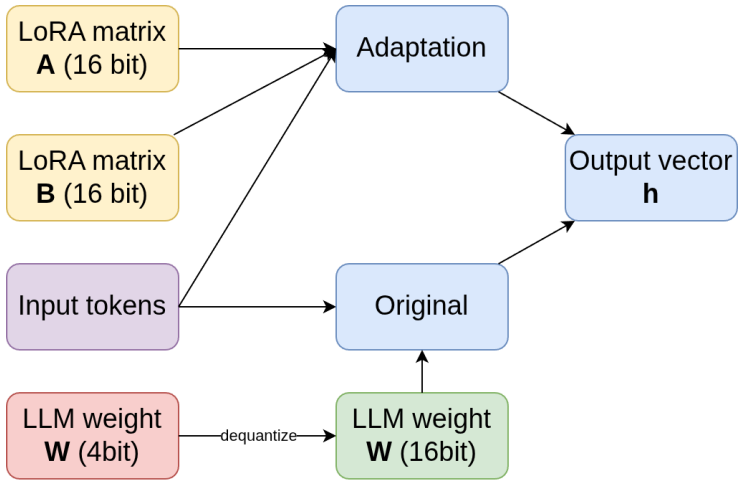
Making finetuning scalable through QLoRA

Subsection 2

How does it differ from LoRA?

QLoRA: Concept

- Load a 4-bit quantized model using a normal approximation of weights.
- Run LoRA in full precision, dequantizing the model weights one at a time.
- Continue as normal.



QLoRA: Algorithm

▷ *====Initialization====* ◁

▷ *Same as before* ◁

▷ *====Backpropagation====* ◁

for all $X \in input\_embeds$ **do**

$\mathbf{\Delta W} = \frac{\alpha}{r}\mathbf{AB}$

$\mathbf{H} = \text{DEQUANT}(\mathbf{W})\ X + \mathbf{\Delta W} X$ ▷ *Output = original + adaptation*

▷ *====End of training====* ◁

▷ *Only attempt dynamic merge. Ignore this advice at your own peril, or if you really know what you are doing.* ◁

QLoRA: Pros and cons

Pros

- Frozen weights no longer monopolize GPU memory and inference time.
- Allows training in consumer-grade hardware.
- Massively reduces training times.

Cons

- End-result is a quantized $\rightarrow$ less powerful model.
- Dequantization imposes a $\approx 10\%$ tax on training times.
- Merging is now much messier.

Summing up

- Full finetuning for LLMs is out of the question.
- However, weight updates are usually low-rank.
- Thus, we can approximate full finetuning with LoRA.
- We may use QLoRA instead to make it much more scalable.